

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



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The ideally exact categories

A non-pointed generalization
of the notion of semiabelian category

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Introduction

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Chapter 1

Theory

1.1 The pullback functor and its adjunction

In this section we will define the pullback functor and its left adjoint, the shrink functor. Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow A$ in $\mathcal{A}$, one can consider adjunction

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xleftarrow{p^*} \\ \top \\ \xrightarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

defined as follows:

- $(\mathcal{A} \downarrow B)$ is the category of objects over B in $\mathcal{A}$, we denote an object by $(A, \alpha : A \rightarrow B)$;
- $(\mathcal{A} \downarrow E)$ is the category of objects over E in $\mathcal{A}$, we denote an object by $(D, \delta : D \rightarrow E)$;
- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \searrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \xrightarrow{\quad \quad \quad} & & \xrightarrow{\quad \quad \quad} & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & \searrow \alpha' & & & & \\ & & B & & & & \\ \text{proj}_1 \nearrow & & \uparrow p & & \nwarrow \text{proj}'_1 & & \\ & & E & & & & \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \swarrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

- the unit η^p

$$\begin{aligned}
 \eta : \text{Id}_{(\mathcal{A} \downarrow E)} &\rightarrow p^* p_! \\
 \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) &\rightarrow (E \times_B D, \text{proj}_1)
 \end{aligned}$$

$$\begin{array}{ccccc}
 D & & \xrightarrow{1_D} & & D \\
 \delta \downarrow & \searrow \exists! \langle \delta, 1_D \rangle & & & \downarrow p\delta \\
 & E \times_B D & \xrightarrow{\quad} & & D \\
 & \downarrow & & & \downarrow p \\
 & E & \xrightarrow{p} & & B
 \end{array}$$

- the counit ϵ^p

$$\begin{aligned}
 \epsilon : p_! p^* &\rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
 \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) &\rightarrow (A, \alpha)
 \end{aligned}$$

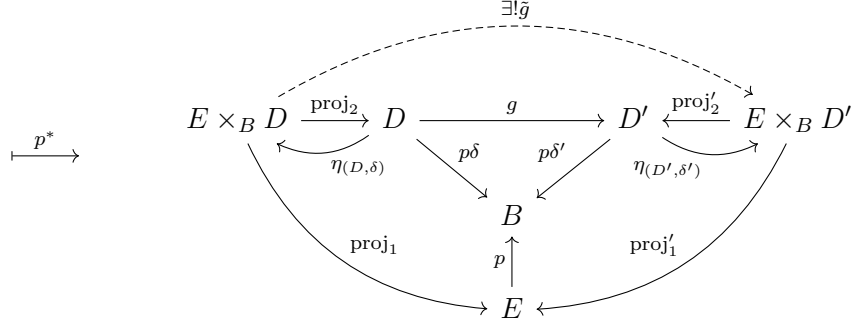
$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2} & A \\
 \downarrow \text{proj}_1 & & \downarrow \alpha \\
 E & \xrightarrow{p} & B
 \end{array}$$

Let us show that this is actually an adjunction. First we want to prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\
 \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\
 D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D'
 \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc}
 D & \xrightarrow{g} & D' \\
 \delta \downarrow & \swarrow \delta' & \\
 E & & \end{array}
 \xrightarrow{p_!}
 \begin{array}{ccc}
 D & \xrightarrow{g} & D' \\
 & \searrow p\delta & \downarrow p\delta' \\
 & & B
 \end{array}$$



Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \text{dashed} & \downarrow p^* p_!(g) = \tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\
 p_! p^*(f) = \tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As similarly done above, we need to expand the action of the functor $p_! p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}_1' \\ & E & & & \end{array} \end{array} \\
 & & \xrightarrow{p!} \\
 & & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ p(\text{proj}_1) \swarrow & & \uparrow & & \searrow p(\text{proj}_1') \\ & & & & \end{array} \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)}$ is the $\text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. So also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^*(\text{proj}_{2,E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{1,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \downarrow \alpha \\ & & B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \text{proj}_{2,E \times_B A} \\ \curvearrowright \\ E \times_B (E \times_B A) \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} E \times_B A \xrightarrow{\text{proj}_{2,E \times_B A}} A \xleftarrow{\text{proj}_{2,E \times_B A}} E \times_B A \\ \quad \quad \quad \downarrow (\alpha)(\text{proj}_{2,E \times_B A}) \quad \quad \quad \downarrow \alpha \\ \quad \quad \quad B \\ \quad \quad \quad \uparrow p \\ \quad \quad \quad E \end{array}
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \exists! \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) & \\
 \text{proj}_{1, E \times_B A} \searrow & E & \xrightarrow{p} & B & \\
 & & & &
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \tilde{\text{proj}}_2 & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A \longleftarrow E \times_B A \\
 & \searrow & & \searrow & \searrow (\alpha)(\text{proj}_{2, E \times_B A}) & \nearrow \alpha & \\
 & & & & B & & \\
 & & & & \uparrow p & & \\
 & & & & E & & \\
 & \searrow \text{proj}_{1, E \times_B A} & & \nearrow \text{proj}_{1, E \times_B (E \times_B A)} & & \nearrow \text{proj}_{1, E \times_B A} & \\
 & & & & & &
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned}
 (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 &= (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = \text{proj}_{2, E \times_B A} = (\text{proj}_{2, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$\begin{aligned}
 (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{1, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 \text{proj}_{1, E \times_B A} &= (\text{proj}_{1, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p_!) \circ (p_! \eta))_{(D, \delta)} &= ((\epsilon p_!)_{(D, \delta)} (p_! \eta))_{(D, \delta)} = (\epsilon_{p_!(D, \delta)} p_!(\langle \delta, 1_D \rangle)) = \\ &= (\epsilon_{(D, p\delta)}) p_!(\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D}(\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds.

$$\begin{array}{ccc} & p^* & \\ (\mathcal{A} \downarrow B) & \top & (\mathcal{A} \downarrow E) \\ & p_! & \end{array}$$

Remark 1.1.1. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

We need the following definition in order to claim some properties.

Definition 1.1.2. Let γ be a natural transformation between two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, we say that γ is a cartesian natural transformation if for every morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram is a pullback:

$$\begin{array}{ccc} F(A) & \xrightarrow{\gamma_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\gamma_B} & G(B) \end{array}$$

ie the naturality diagrams are pullback.

Property 1.1.3. The unit η and the counit ϵ are cartesian.

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Considering the commutative diagram below, since both the bottom and the total diagrams are pullback, it follows that the top diagram is a pullback.

$$\begin{array}{ccccc} E \times_B D & \xrightarrow{\text{proj}_2} & D & & \\ \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g & & \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' & & \\ \downarrow \text{proj}'_1 & & \downarrow p\delta' & & \\ E & \xrightarrow{p} & B & & \end{array} \quad \begin{array}{l} \text{proj}_1 \swarrow \quad \searrow p\delta \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc}
 & & 1_D & & \\
 & \swarrow & \text{---} & \searrow & \\
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\
 \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & \nwarrow & \text{---} & \swarrow & \\
 & & 1_{D'} & &
 \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. Considering the following commutative diagram, since the bottom diagram is a pullback and the total diagram is a pullback, we can conclude that the top diagram is a pullback, as stated.

$$\begin{array}{ccccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)} = \text{proj}_2} & A & & \\
 \downarrow p_! p^*(f) = \tilde{f} & & \downarrow f & & \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')} = \text{proj}'_2} & A' & & \\
 \downarrow \text{proj}'_1 & & \downarrow \alpha' & & \\
 E & \xrightarrow{p} & B & &
 \end{array}$$

proj_1 (left curved arrow), α (right curved arrow)

□

1.2 The monad associated to the adjunction

Consider the monad associated to the adjunction we just studied, we want to study some properties that will be useful. In order to do that we need first to define some notions.

Definition 1.2.1. A *connected limit* is a limit over a connected category. A category is *connected* if it is inhabited and any two objects can be connected by a zigzag of morphisms

$$\begin{array}{ccccccc}
 x_0 = y_1 & & & y_3 & & & y_n = x_1 \\
 & \swarrow & & \swarrow & \searrow & & \swarrow \\
 & & y_2 & & \dots & &
 \end{array}$$

where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 1.2.2. Some examples

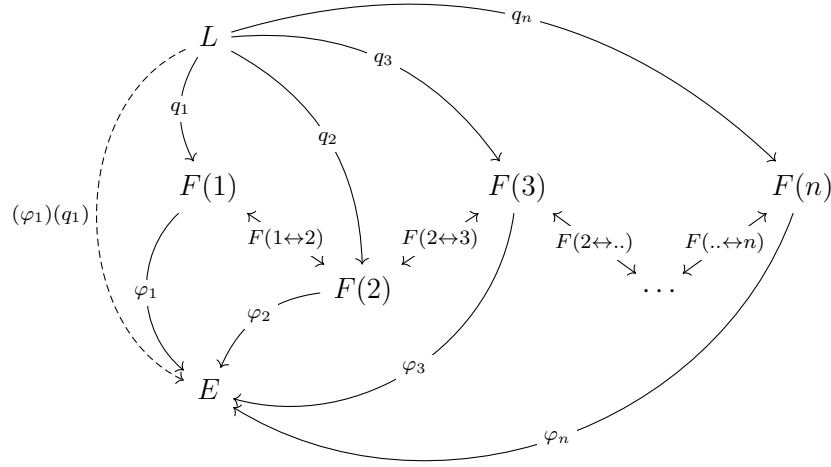
- The pullback is a connected limit.

$$\begin{array}{ccc}
 & 2 & \\
 & \downarrow j & \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$1 \begin{matrix} \xrightarrow{i} \\ \xrightarrow{j} \end{matrix} 2$$

Remark 1.2.3. The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. In fact we can provide to the objects in $\mathcal{A}$ a morphism to form an object in $(\mathcal{A} \downarrow E)$. In the zigzag below we chose for example $(\varphi_1)(q_1)$, but we could use whatever morphism of the type $(\varphi_i)(q_i)$. The connection guarantees that all the morphisms $q_1 \dots q_n$ involved in the limit are also morphisms in $(\mathcal{A} \downarrow E)$, ie $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i \in 1, \dots, n$.



Remark 1.2.4. The product is not a connected limit, in fact we can see that the following category is not connected.

$$1 \quad 2$$

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ') be two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccc} D \times_E D' & \xrightarrow{\quad} & D' \\ \swarrow & & \searrow \delta' \\ D & \xrightarrow{\delta} & E \\ \searrow \delta & & \parallel \\ & & E \end{array}$$

which doesn't correspond to the product $D \times D$ in $\mathcal{A}$.

Property 1.2.5. The monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^* \epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullback, since pullbacks are connected limit and hence are computed in $\mathcal{A}$, which has pullbacks. The monad $p^* p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks because they are computed in $\mathcal{A}$ and $p_!$ doesn't affect the object and morphism in $\mathcal{A}$. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^* \epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. $\square$

Definition 1.2.6. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be *essentially nullary* if the morphism $[T(!_X), \eta_X]$ is a strong epimorphism for each object X in $\mathcal{X}$, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X .

Now we claim a lemma which will be useful to prove some theorems.

Lemma 1.2.7. A category $\mathcal{X}$ is protomodular id and only if for each pullback

$$\begin{array}{ccc} B' \times_B A & \xrightarrow{k} & A \\ \downarrow \uparrow & & \downarrow \uparrow \\ B' & \xrightarrow{\beta} & B \end{array} \quad \begin{array}{c} p \\ s \end{array}$$

the pair (k, s) is strongly epimorphic.

The proof is out of the scope of this thesis, but we can find it in (citazione)

Theorem 1.2.8. If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.

Proof. For each object X in $\mathcal{X}$ we can consider the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ !^X \downarrow \uparrow !_X & & T(!^X) \downarrow \uparrow T(!_X) \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array}$$

Since $\mathcal{X}$ is protomodular then by the lemma above, the pair $(\eta_X, T(!_X))$ is a strongly epimorphic. It is a standard fact that given f and g morphism, if (f, g) is a strongly epimorphic pair, then $[f, g]$ is a strong epi. So we conclude that $[\eta_X, T(!_X)]$ is a strong epi, ie the monad is essentially nullary. (citazione) $\square$